Eine Woche, ein Beispiel

4.19. branching rules

Ref:
[HTW05]: Roger Howe, Eng-Chye Tan, and Jeb Willenbring. "Stable Branching Rules for Classical Symmetric Pairs." Transactions of the American Mathematical Society 357, no. 4 (2005): 1601–26. https://doi.org/10.1090/S0002-9947-04-03722-5.

Here are just some cartoons verifying the formulas.

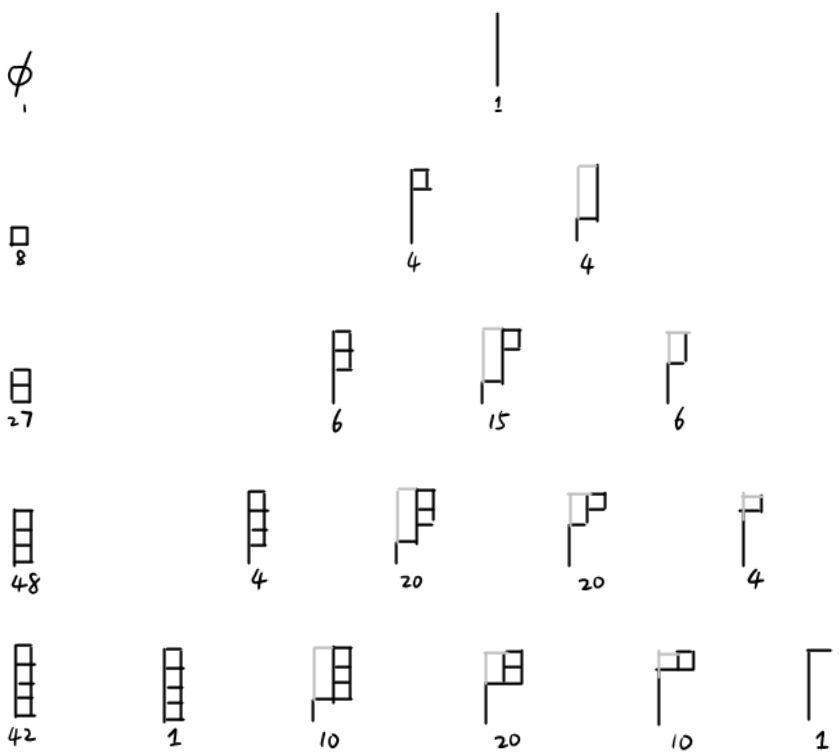
[HTW05, Thm 1.2]

$$GL(2n;\mathbb{C}) \longleftarrow\!\!\!

[HTW05, 2.3.2]

$$Sp(2n;\mathbb{C}) \longleftarrow GL(n;\mathbb{C})$$

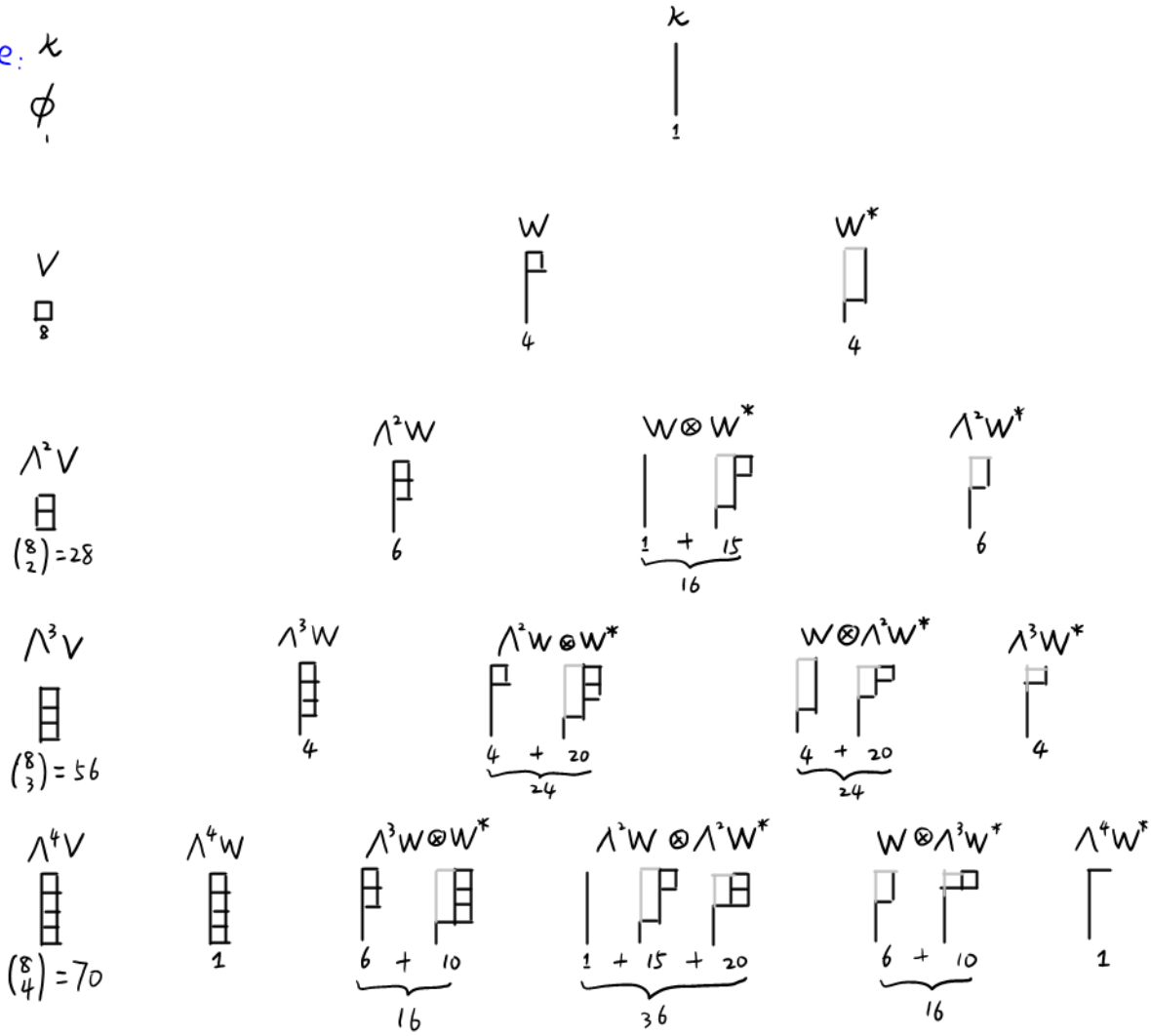
$n=4$ case:



Combining them, we get:

$$GL(2n;\mathbb{C}) \hookleftarrow Sp(2n;\mathbb{C}) \hookleftarrow GL(n;\mathbb{C})$$

$n=4$ case:



Using the same method, we can compute:

$$GL(2n;\mathbb{C}) \longleftarrow Sp(2n;\mathbb{C}) \longleftarrow GL(n;\mathbb{C})$$

$n=4$ case:

k $\qquad\qquad$ k

ϕ $\qquad\qquad$ 1

V $\qquad\qquad$ W $\qquad\qquad$ W^*

8 $\qquad\qquad$ 4 $\qquad\qquad$ 4

$\mathrm{Sym}^2 V$ $\qquad\qquad$ $\mathrm{Sym}^2 W$ $\qquad\qquad$ $W \otimes W^*$ $\qquad\qquad$ $\mathrm{Sym}^2 W^*$

$\binom{9}{2} = 36$ $\qquad\qquad$ 10 $\qquad\qquad$ $\underbrace{1 + 15}_{16}$ $\qquad\qquad$ 10

$\mathrm{Sym}^3 V$ $\qquad\qquad$ $\mathrm{Sym}^3 W$ $\qquad\qquad$ $\mathrm{Sym}^2 W \otimes W^*$ $\qquad\qquad$ $W \otimes \mathrm{Sym}^2 W^*$ $\qquad\qquad$ $\mathrm{Sym}^3 W^*$

$\binom{10}{3} = 120$ $\qquad\qquad$ 20 $\qquad\qquad$ $\underbrace{4 + 36}_{40}$ $\qquad\qquad$ $\underbrace{4 + 36}_{40}$ $\qquad\qquad$ 20

$\mathrm{Sym}^4 V$ $\qquad\qquad$ $\mathrm{Sym}^4 W$ $\qquad\qquad$ $\mathrm{Sym}^3 W \otimes W^*$ $\qquad\qquad$ $\mathrm{Sym}^2 W \otimes \mathrm{Sym}^2 W^*$ $\qquad\qquad$ $W \otimes \mathrm{Sym}^3 W^*$ $\qquad\qquad$ $\mathrm{Sym}^4 W^*$

$\binom{11}{4} = 330$ $\qquad\qquad$ 35 $\qquad\qquad$ $\underbrace{10 + 70}_{80}$ $\qquad\qquad$ $\underbrace{1 + 15 + 84}_{100}$ $\qquad\qquad$ $\underbrace{10 + 70}_{80}$ $\qquad\qquad$ 35

168 $\qquad\qquad$ 20 $\qquad\qquad$ $\underbrace{2 \cdot 4 + 36}_{44}$ $\qquad\qquad$ $\underbrace{2 \cdot 4 + 36}_{44}$ $\qquad\qquad$ 20